

Chapter 4

Windows and Linux Resource Monitoring and Management

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Outlines

- *Resource Monitoring & Management I*
- Stand-alone systems (Memory, Disk Use, CPU Use)
- Monitoring Tools
- Routine Resource Maintenance
- *Resource Monitoring & Management II*
- Bandwidth
- *File & Mail Servers*
- Remote Administration
- Performance
- RAID
- SCSI

Resource Monitoring and Management

Resource Monitoring and Management (RMM)

- It is the process of **tracking** and **optimizing** the use of **resources** within a system or application.
- This can include monitoring the usage of **CPU**, **memory**, **disk space**, **network bandwidth** and **other system resources** to ensure that they are being used efficiently and effectively.
- **Resource monitoring** can help to identify potential performance issues or bottlenecks before they become critical, and enable administrators to take proactive measures to optimize resource allocation and prevent downtime.

Resource Monitoring and Management ...

Windows Resource Monitoring includes **five** tabs:

- 1. Overview:** contains a complete list of **processes running** on the system
- 2. CPU:** displays column lists of Processes, Services, Associated Handles and Associated Modules; charts of CPU Usage (separate for every core)
- 3. Memory:** displays overall Physical Memory consumption and separate consumption of every Process; charts of Used Physical Memory, Commit Charge and Hard Faults/sec
- 4. Disk:** displays Processes with Disk Activity, Disk Activity and Storage; charts of Disk Usage (KB/sec) and Disk Queue Lengths

Resource Monitoring and Management ...

Window Resource Monitoring includes five tabs...

- 5. Network:** displays Processes with Network Activity, Network Activity, TCP Connections and Listening Ports; charts of Network Usage (separate for every adapter) and TCP Connections resource monitoring aspect of the tool: CPU, Memory, Disk, and Network

Resource Monitoring and Management ...

All program → Control Panel → System and Security → Administrative Tools → Resource monitoring

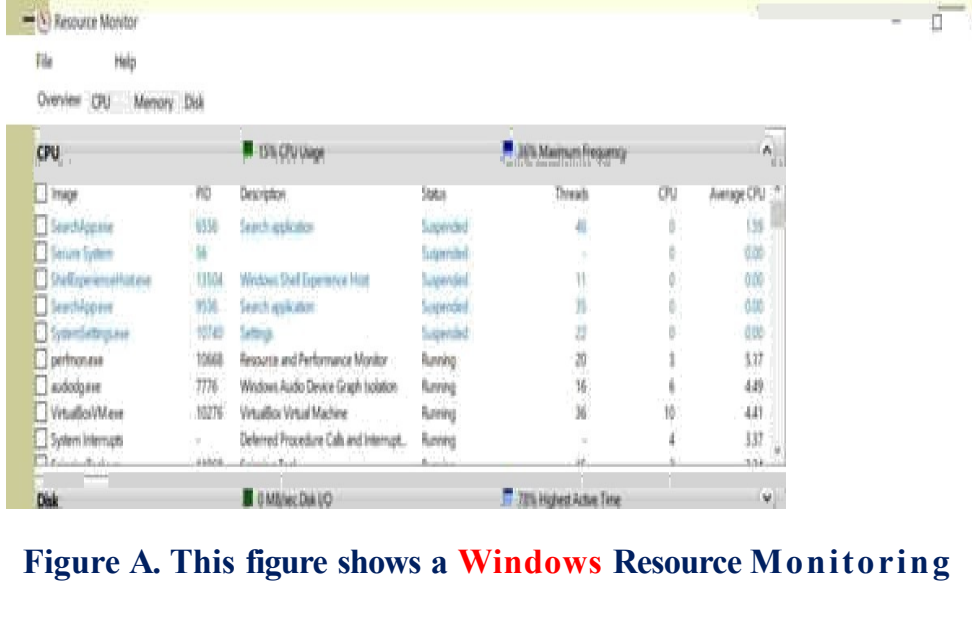


Figure A. This figure shows a **Windows** Resource Monitoring

Resource Monitoring and Management

Processes with Disk Activity:

- When monitoring windows processes with disk activity, a list of all processes that are using disk resources is displayed
 1. **Image:** the name of the process executable file.
 2. **PID:** the process ID number associated with the process
 3. **Read (B/sec):** average number of bytes read per second by the process in the past minute.
 4. **Write (B/sec):** average number of bytes written per second by the process in the past minute.
 5. **Total (B/sec):** average number of bytes accessed per second (both read and write in the past minute.

Resource Monitoring and Management ...

Disk Activity

- *Is **read** and **write** operations that occur on a computer's hard disk or storage device.*
- *When monitoring disk activity, some disk monitoring tools may show information like:*
- **File:** *name of the file that is being used by the active process.*
- **I/O Priority:** *the priority of I/O transfers that can be used to prioritize disk access for certain processes*
- **Response Time (ms):** *the disk response time in milliseconds*
- *This can be used to identify potential bottleneck issues related to disk access time*

Resource Monitoring and Management ...

Storage

- **Logical Disk:** The drive letter associated with a disk
- **Physical Disk:** Which physical disk is being monitored on this line?
- **Active Time (%)** Shows the percentage of time that the **disk is not idle** and is actively serving requests. If a disk is constantly turning at very high levels (say, more than 80%).
- **Available Space (MB):** How much disk space is available on volume?
- **Total Space (MB):** What is the total size of the volumes
- **Disk Queue Length:** average disk queue length.
- **The Queue length:** metric displays the number of outstanding requests (read and write) at any given time.

Resource Monitoring and Management

Network Activity

- It refers to the **exchange of data between devices on a network**, including **incoming and outgoing data packets**.
- It can be monitored and analyzed using network monitoring tools
- Provide insight into network performance, security, and usage.
- **Network I/O:** to transfer data between a computer or device and network
- **Network Utilization:** The amount of network bandwidth that is being used at any given time
- **Address:** IP or MAC address

Resource Monitoring and Management ...

Network Activity ...

TCP Connections

- **Local Address:** The IP address of the device that initiating a network connection
- **Local Port:** A network port number used by the device that initiating a network connection
- **Remote Address:** The IP address of the device that receiving data from a network connection.
- **Remote Port:** a network port number used by computer that receiving data from a network connection.
- **Packet Loss:** the percentage of data packets that are lost during transmission over a network.

Resource Monitoring and Management ...

Network Activity ...

Listening Ports: *a network port on a computer or device that is open and waiting for incoming network connections*

- **Address:** *Refers to an IP address or a MAC address.*
- **Protocol:** *Is a **set of rules and procedures** for transmitting data between devices on a network*
- **Firewall:** *used to monitor and control incoming and outgoing network traffic **based on predefined security rules***
- **Status:** *The current state of a network connection or service.*
- **E.g. Service may be running or stopped, a network connection may be established, or disconnected, or a firewall rule may be enabled or disabled.**

Resource Monitoring and Management ...

Windows servers Monitoring Tools

- Several monitoring tools available for Windows servers used to monitor resource usage and performance
- **PRTG Network Monitor**: A comprehensive network monitoring tool that used to monitor Windows Server resources such as **CPU usage, memory, disk space, and network utilization**.
- It provides real-time alerts and detailed performance reports.
- **SolarWinds Server & Application Monitor**: A monitoring tool that used to monitor windows server resources, applications, and services; provides real-time alerts, performance metrics and detailed reporting

Resource Monitoring and Management ...

Network System Monitoring Tools

- **Zabbix:** It provides real-time monitoring, alerting and visualization of system performance data.
- **Prometheus:** An open-source monitoring and alerting toolkit that specializes in time-series data and can monitor system resources, applications, and network performance.
- **Nagios:** A popular open-source system monitoring tool that can monitor a wide range of system resources, including servers, applications, and network devices.

Group Assignment

Direction: Get yourselves in a group of 5 or 6 members and work on the following group work task. AI generated and/or copied work will have poor mark.
(5 Pts)

Group Work Task: (10 Pages)

This time we as a network system administrators have different **Network System Monitoring Tools. Report in detail the different types of Network System Monitoring Tools.** As a network system administrators which monitoring tool you choose to work with? Why?

Routine Resource Maintenance

- It is the regular upkeep of system resources to ensure that they are operating efficiently and effectively.
- **Regular activities** of system administrator can include tasks such as:
- **Disk clean up:** Cleaning up unnecessary files and folders from system
- **Defragmentation:** help optimize disk usage and improve system performance.
- **Software updates:** Keeping the operating system and software applications up-to-date
- **Security updates**
- **Regular backups**
- **Hardware maintenance:** fans, power supplies, and cooling systems

Resource monitoring and Management II

Bandwidth: the **maximum amount of data** that is transmitted over a network or internet connection in a given amount of time

- Measured in bits per second (bps), Kbps, Mbps, or Gbps.
- It is a critical factor in determining the **speed** and **efficiency** of data transmission.
- Determines how quickly data can be transferred between devices on a network or between a device and the internet.

Resource Monitoring And Management II ...

A file server: *It is a server that dedicated to **storing and managing files** that accessed by authorized users on a network.*

Types of File Servers:

- *A file server may be dedicated or non-dedicated.*
- **A dedicated file server:** *is designed specifically for use as a file server, with workstations attached for reading and writing files and databases.*
 - *The practice of having a unique IP and specific server serving only specific user*
- **A Non dedicated file server:** *It is for general use as a file server*
 - *A number of websites and people can benefit from the same server,*

Resource Monitoring and Management

- **Network printers:** *are printers that are connected to a network and can be accessed by multiple users from different devices.*
- **Remote administration:** *is the process of managing and controlling a computer or network from a remote location.*
- *Enables administrators to manage systems and networks from a central location, which can increase efficiency and reduce costs*
- **Benefits of remote administration, include:**
 - *Increased efficiency*
 - *Cost savings:*
 - *Improved security*

Linux Resource Monitoring & Management

- Linux provides several tools for monitoring and managing system resources, such as CPU, memory, disk, and network usage
- Some of the key tools and techniques for Linux Resource Monitoring and Management are:

1) Monitoring Tools

- Linux provides several tools for monitoring system resources
- A) **top**: a command-line tool that displays real-time information about system processes, including CPU and memory usage.
- B) **htop**: a more advanced version of top that provides additional features and a more graphical user interface.
- C) **nmon**: a command-line tool that displays real-time information about system performance, including CPU, memory, disk, and network usage

Linux Resource Monitoring & Management ...

1) Monitoring Tools ...

- D) **Iostat**: A command-line tool that displays information about disk I/O performance
- E) **Sar**: A command-line tool that collects system performance data over time and generates reports.

2) Network Bandwidth and Processing Power

- To optimize network bandwidth and processing power, you can:
 - A) **Use a high-quality network equipment** and optimize network configuration to maximize network bandwidth.
 - B) **Use a high-quality processors** and optimize hardware and software configurations to maximize processing power.
 - C) **Use load balancing** and other techniques to distribute processing load across multiple systems.

Linux Resource Monitoring & Management ...

3) Managing Storage

- Linux involves several tasks, including creating partitions, formatting file systems, and monitoring and mounting storage devices
- In Linux systems, to effectively manage storage, you can:
 - a. Use tools like **fdisk** or **parted** to create partitions.
 - b. Use tools like **mkfs** or **mkfs.ext4** to format file systems.
 - c. Use the **mount** and **umount** commands to mount and unmount storage devices.
 - d. Use tools like **fdisk**, **parted**, and **lsblk** to manage storage devices and file systems.

Performance

- *It is the speed and efficiency with which software performs its intended work*
- **Algorithm design:** Well-designed algorithms can improve the speed and efficiency of software.
- **Code optimization:** reducing the time and resources required to execute code.
- **Hardware resources:** fastest processors, more memory, and large Storage can all improve software performance
- **Network performance:** rely on network communication.

Redundant Array of Independent Disks (RAID)

(Reading Assignment)

- It is a technology that allows **multiple hard drives** to be used together to **improve performance, reliability, and/or storage capacity**.
- It is used in **servers and storage devices** to provide **fault tolerance** and improve performance.
- There are **several RAID levels**, each with its own characteristics:
- **RAID 0**
 - It uses **striping** to spread data across multiple drives, which can **improve performance**
 - It allows the data to be read and written to multiple drives **simultaneously**.



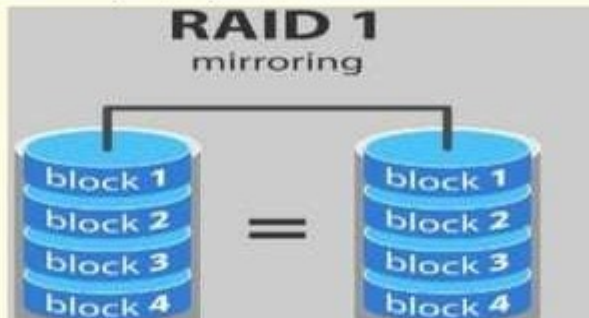
Redundant Array of Independent Disks(RAID) ...

- ❑ **Drawback of RAID 0:** *it does not provide redundancy, so if one drive fails, all data is lost.*

❑ RAID 1

- ✓ It uses **mirroring** to create identical copies of data on two drives.
- ✓ Provides redundancy, if one drive fails, the other can continue to function.

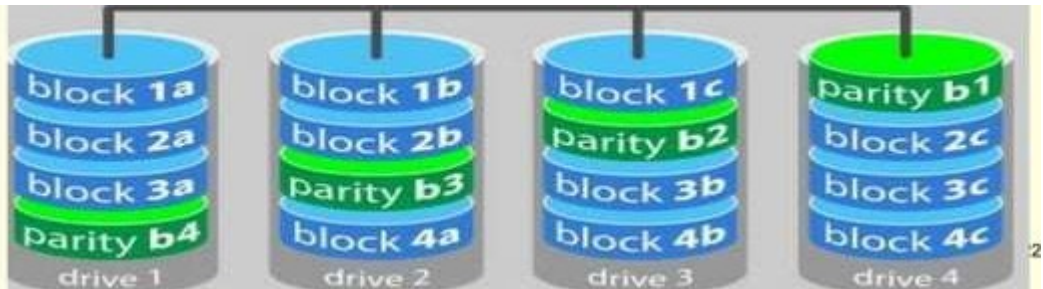
- ❑ **Drawback :** Does not improve performance as data is only written to one drive at a time



Redundant Array of Independent Disks (RAID) ...

• RAID 5

- It uses striping with **parity** to improve performance and provide redundancy
- Data is spread across multiple drives, and parity information is stored on each drive
- If one drive fails, missing data can be reconstructed using the **parity information** on the remaining drives and requires at least **three** drives

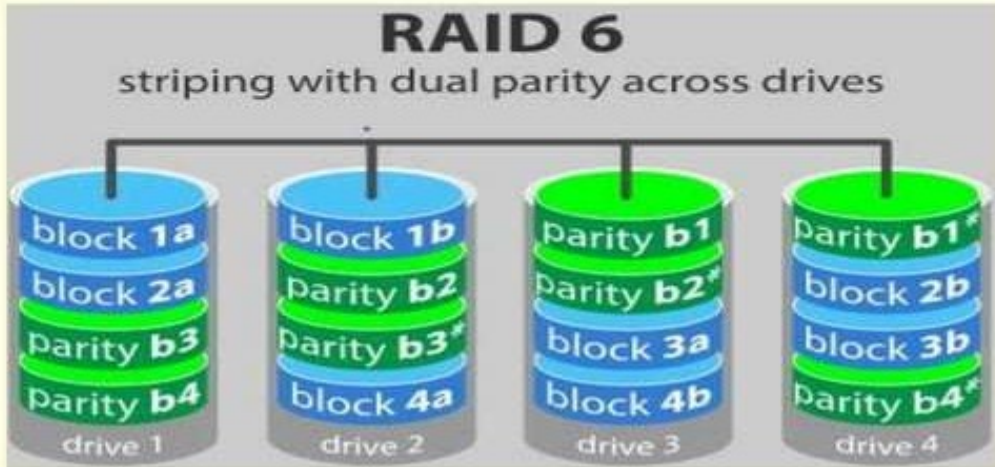


Redundant Array of Independent Disks (RAID) ...

RAID 6

It is similar to RAID 5, but **uses two sets of parity** information to provide **redundancy** to increase recovery and performance.

3 It requires at least **four** drives.



Redundant Array of Independent Disks(RAID) ...

RAID 10

Combines **mirroring** and **striping** to provide both performance and redundancy.

- Data is mirrored across two sets of drives and striped across those sets
- It requires at least **four** drives.



Small Computer Systems Interface (SCSI) (Reading Assignment)

- **Many versions**

- SCSI-1 (1986) 8-bits, 5MB/s
- SCSI-2 (1990) added command queuing, ...
- Fast SCSI-2 8-bits, 10MB/s
- Fast/wide SCSI-2 16-bits, 20MB/s
- Ultra SCSI 8 bits, 20MB/s
- Wide Ultra SCSI 16bits, 40MB/s
- Wide Ultra2 SCSI 16bits, 80MB/s
- Wide Ultra3 SCSI 16bits, 160MB/s
- Ultra-320, Ultra-640 SCSI



Small Computer System Interface (SCSI) ...

- It is a set of **standards** for connecting and transferring data between computers and peripheral devices, such as hard drives, tape drives and scanners.
- It is used in **server** and **storage environments**, where high performance and reliability are critical
- There are several types of SCSI interfaces, including:
 - **SCSI-I**
- It was the first SCSI standard, introduced in 1986.
- It supports data transfer rates of up to 5 MB/s and allows for up to 8 devices to be connected to a single bus.

Small Computer System Interface (SCSI) ...

■ SCSI-2

- It was introduced in 1990 and added new features, such as disconnect/reconnect and synchronous data transfer.
- It also increased the maximum data transfer rate to 10 MB/s and allowed for up to 16 devices to be connected to a single bus.

■ SCSI-3

- It is the recent SCSI standard, introduced in 1996.
- It added several new features, such as packetized data transfer and support for Fiber Channel and Serial Attached SCSI (SAS) interfaces
- It also increased the maximum data transfer rate to 80 MB/s and allowed for up to 32 devices to be connected to a single bus.

Small Computer System Interface(SCSI) ...

- **SCSI** interfaces offer several advantages over other interfaces, such as IDE and SATA including:
 - **High performance**: support high data transfer rates and low latency, which makes them ideal for **Server** and **storage** environments.
 - **Scalability**: support multiple devices on a single bus, which makes them easy to scale as storage needs grow.
 - **Reliability**: use a robust protocol and are designed for enterprise environments, which makes them more reliable than consumer-grade interfaces

End of Chapter 4

Thank you !

Any Questions, Please